

# Subscription Tracker Dashboard

## Project Overview

Subscription Tracker Dashboard is a full-stack web application developed using React.js and Node.js. The application helps users manage their digital subscriptions by tracking recurring expenses, monitoring renewal dates, and calculating monthly spending.

The system provides a centralized dashboard that allows users to add, view, update, pause, and delete subscriptions while displaying key metrics such as monthly burn rate and upcoming renewals.

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## Problem Statement

Build a Subscription Tracker Dashboard that enables users to:

- Add subscription services.
- Track recurring monthly and yearly payments.
- Calculate total monthly spending.
- Identify subscriptions renewing soon.
- Manage subscription status (Active/Paused).
- Delete subscriptions when no longer needed.

All calculations and business logic must be handled on the backend.

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## Technology Stack

### Frontend

- React.js
- Axios
- CSS3

### Backend

- Node.js
  - Express.js
  - UUID
  - CORS
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# System Architecture

User ↓ React Frontend ↓ REST API Calls ↓ Node.js + Express Backend ↓ Business Logic Layer ↓ In-Memory Data Store

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## Features

### 1. Add Subscription

Users can create a subscription by providing:

- Service Name
  - Cost
  - Billing Cycle (Monthly/Yearly)
  - Next Renewal Date
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### 2. Dashboard Metrics

#### Monthly Burn Rate

Represents the total monthly spending across all active subscriptions.

Formula:

Monthly Burn Rate = Sum of Monthly Rates

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#### Upcoming Renewals

Displays the count of active subscriptions renewing within the next 7 days.

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### 3. Subscription Management

Users can:

- View subscriptions
  - Pause subscriptions
  - Reactivate subscriptions
  - Delete subscriptions
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## 4. Renewal Alerts

Subscriptions renewing within 7 days are marked with a:

"Renewing Soon" badge

This helps users avoid unexpected charges.

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## Backend Business Logic

### Cost Normalization Engine

Converts all subscription costs into a monthly equivalent.

Examples:

Monthly Subscription: Cost = ₹500

Monthly Rate = ₹500

Yearly Subscription: Cost = ₹1200

Monthly Rate = ₹100

Formula:

Monthly Rate = Yearly Cost ÷ 12

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### Renewal Date Calculator

Calculates the remaining days before renewal.

Formula:

Days Remaining = Renewal Date - Current Date

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### Renewal Alert Engine

If:

$0 \leq \text{Days Remaining} \leq 7$

Then:

Renewing Soon = True

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## API Endpoints

### Get All Subscriptions

GET /api/subscriptions

Response:

- Subscription List
  - Dashboard Metrics
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### Add Subscription

POST /api/subscriptions

Creates a new subscription.

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### Toggle Status

PATCH /api/subscriptions/:id/toggle

Changes:

Active → Paused

Paused → Active

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### Delete Subscription

DELETE /api/subscriptions/:id

Removes a subscription.

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## Get Metrics

GET /api/metrics

Returns:

- Monthly Burn Rate
  - Upcoming Renewal Count
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## Frontend Components

### Header Component

Displays application title and dashboard header.

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### Metrics Component

Displays:

- Monthly Burn Rate
  - Upcoming Renewals
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### Subscription Form Component

Handles:

- User input
  - Form validation
  - API submission
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### Subscription Table Component

Displays all subscription records.

Supports:

- Toggle Status
  - Delete Subscription
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## Toast Component

Displays success notifications for:

- Add
  - Update
  - Delete actions
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## Folder Structure

frontend/

- src/
- components/
  - Header.jsx
  - Metrics.jsx
  - SubscriptionForm.jsx
  - SubscriptionTable.jsx
  - Toast.jsx
- services/
  - api.js
- App.jsx
- App.css
- main.jsx

backend/

- controllers/
  - routes/
  - services/
  - utils/
  - data/
  - server.js
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## Assumptions

- Data is stored in memory.
  - No database integration.
  - Active subscriptions only contribute to burn rate.
  - Renewal alerts are generated within 7 days.
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## Future Enhancements

- MongoDB Integration
  - User Authentication
  - Email Notifications
  - Payment Gateway Integration
  - Cloud Deployment
  - Mobile Responsive UI
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## Conclusion

The Subscription Tracker Dashboard provides an efficient solution for managing recurring subscriptions. The application demonstrates full-stack development concepts including REST API design, React component architecture, backend business logic implementation, and real-time dashboard analytics.